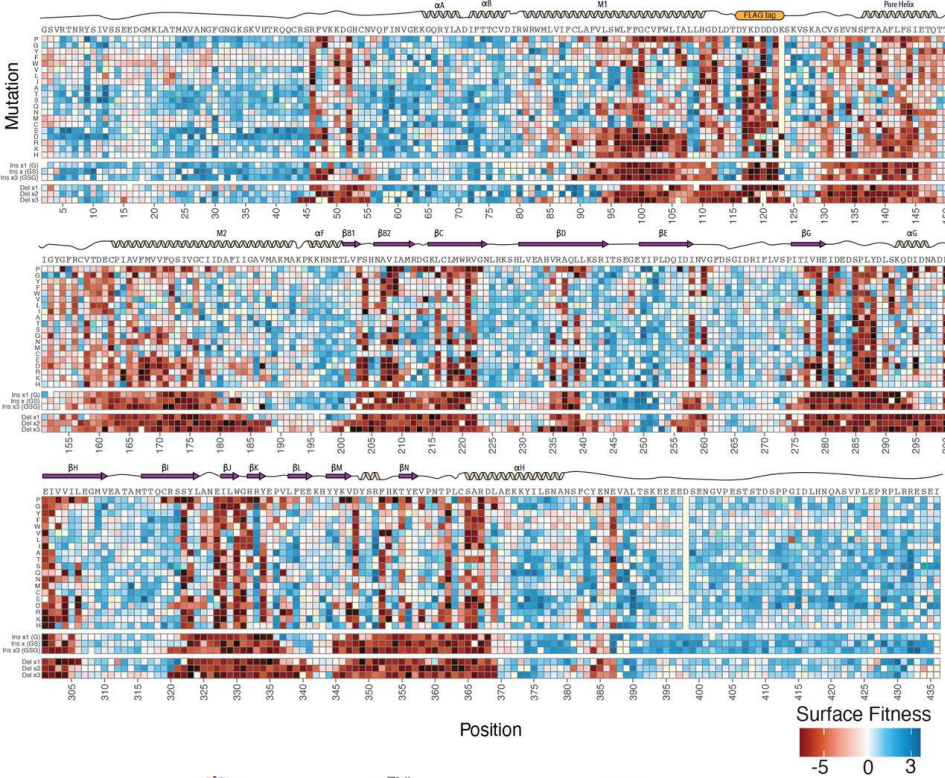


An Introduction to Probabilistic Model Building or thinking about models in the era of AI

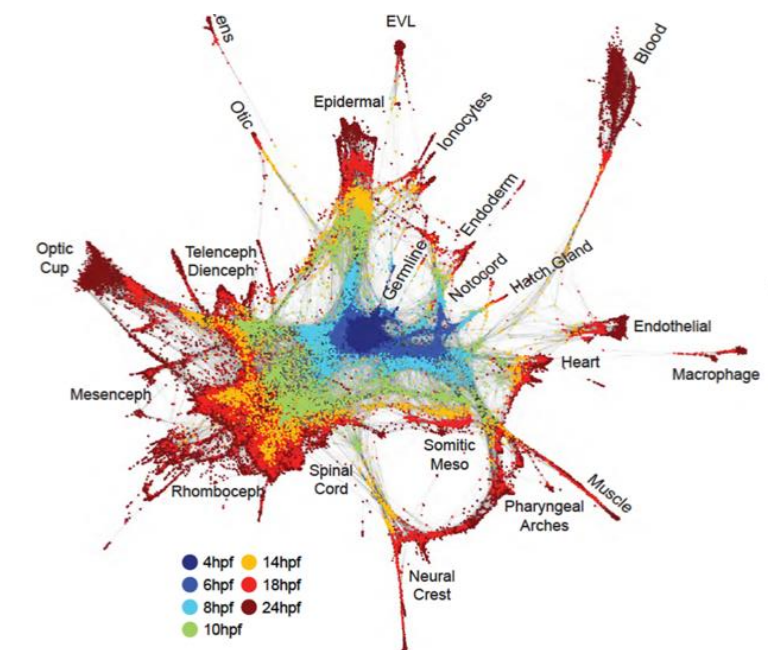
Mafalda Dias & Jonathan Frazer

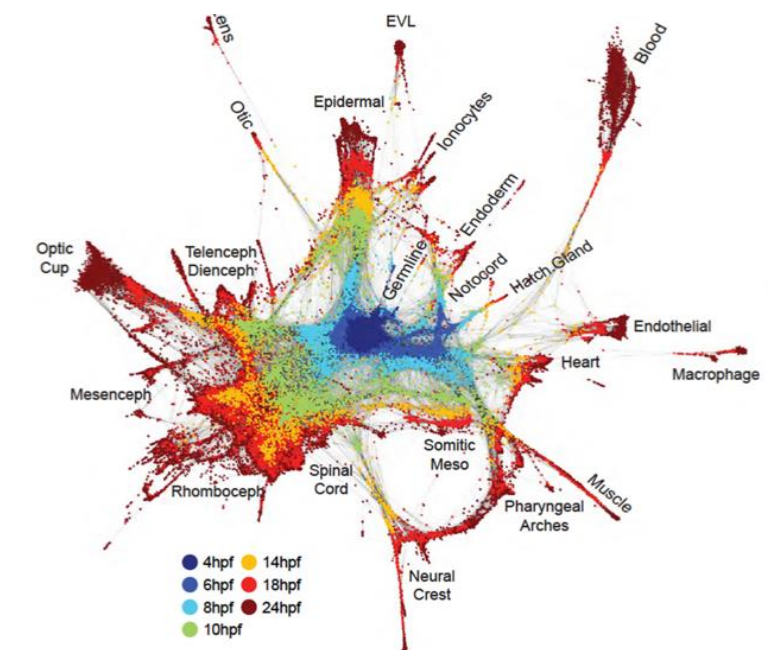
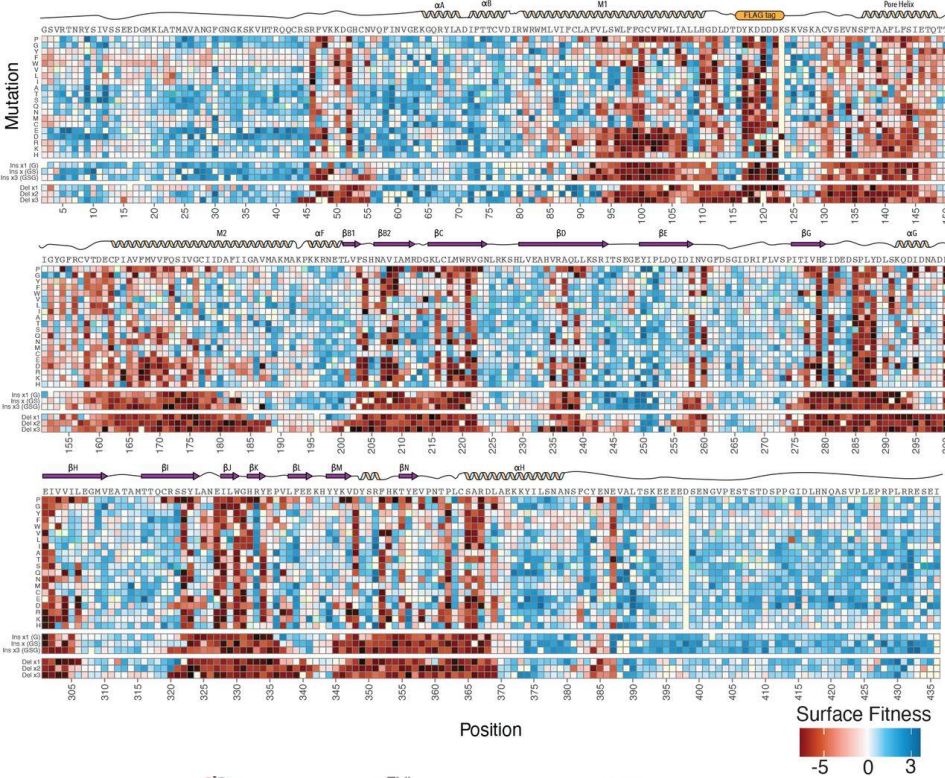


The ever growing need for probabilistic model building in biology

Data getting more quantitative, complex and larger – how to distill this information?

LLMs generate entire code bases, train models, run experiments – more time to think about model building assumptions and how this relates to biology.





The ever growing need for probabilistic model building in biology

Data getting more quantitative, complex and larger – how to distill this information?

LLMs generate entire code bases, train models, run experiments – more time to think about model building assumptions and how this relates to biology.

We will introduce you to the beautiful and intimately linked worlds of **generative** and **Bayesian** modelling – a framework for:

- thinking about **beliefs**
- how they should change as a **result** of new **observations**
- how to **design efficient experiments**

Session 1: Probabilistic modelling (a Bayesian perspective)

Bayesian models as generative stories

Inductive bias: encoding what you know

Smoothness and sparsity

Notes on inference

Session 2: Latent variable models

Hierarchical models as latent variable models

Variational Autoencoders and ELBO

From VAEs to Diffusion

Session 3: Computer aided experimental design

Bayesian Experimental Design

Compressed sensing

What we will not cover:

Workflow – testing if training went well or not, if the model fits the data well, iteration and expectations of a model more generally.

Inference – the other side of the coin to modelling.

Finetuning and reinforcement learning of foundation models – building models on top of general models.

Many other probabilistic modelling approaches – non-parametric (i.e. gaussian processes), attention, normalizing flows, etc etc etc

Session 1: Probabilistic modelling (a Bayesian perspective)

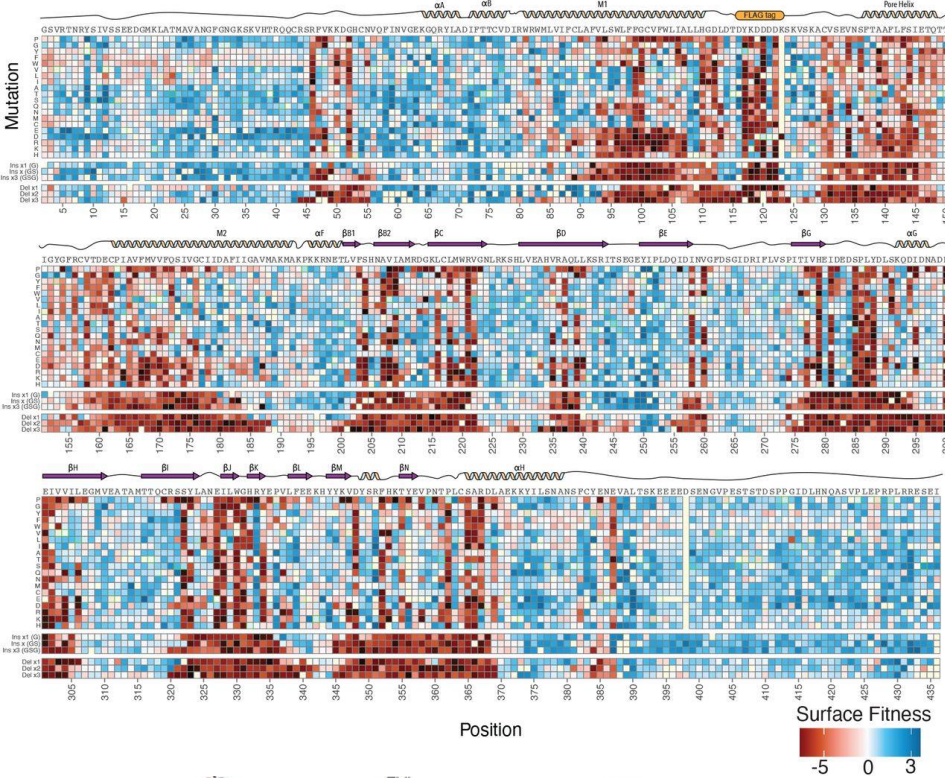
Bayesian models as generative stories
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Smoothness and sparsity
Notes on inference

What does it mean to "model" something?

A joint distribution over observables.

A very Bayesian perspective: characterise data by characterising the underlying model or distribution.

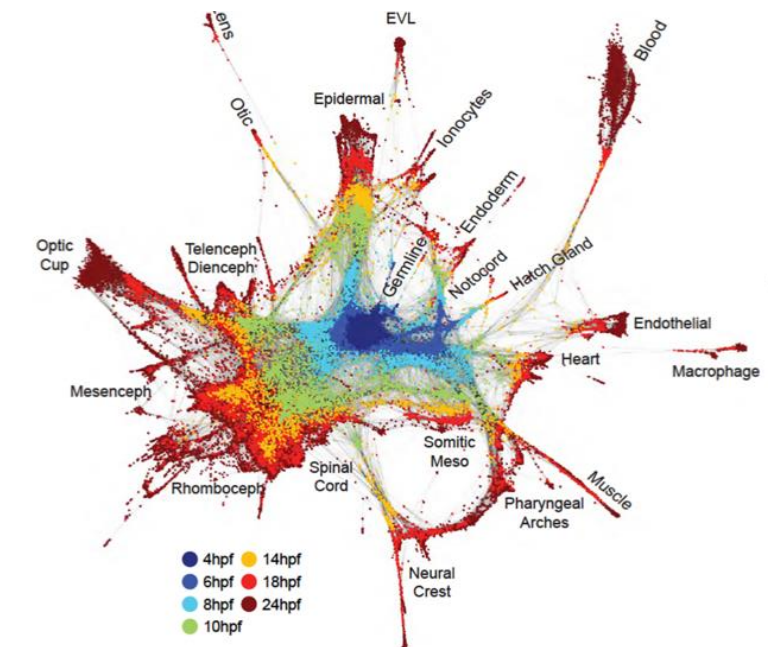
To make a model you have to make simplifying assumptions. The art is in making **simplifying assumptions that preserve key features of reality**, thus enabling us to learn something from the model and/or use it for something.

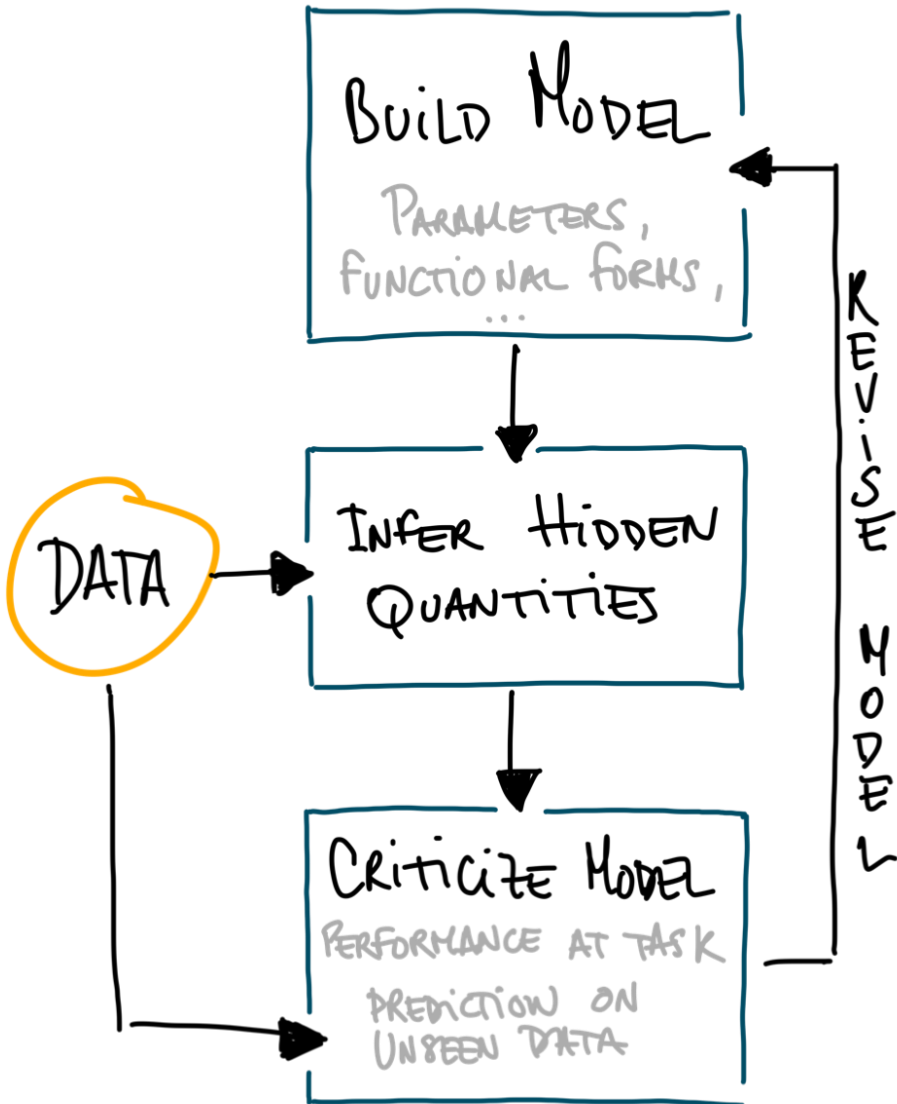


What does it mean to "model" something?
A joint distribution over observables.

Pretty much whenever we design an experiment, or analyse data, and certainly if we make any kind of hypothesis, then implicitly or otherwise, we are building a model.

Thinking about model building is a way to make our assumptions explicit, so that they can be challenged, improved, and combined with domain knowledge.





What does it mean to "model" something?
A joint distribution over observables.

Thinking about model building is a way to **make our assumptions explicit**, so that they can be challenged, improved, and combined with domain knowledge.

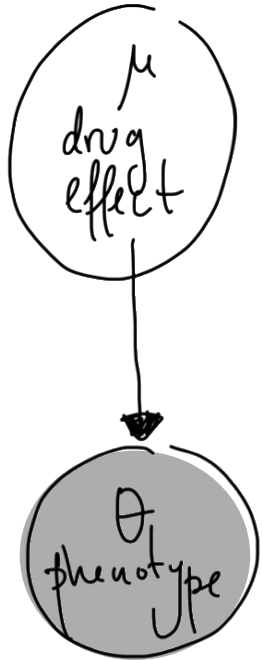
Box's loop – A modern interpretation of Box's perspective (1976) on the process of **model building** – inference – validation – revision.

Telling Generative Stories

In recent years there has been a lot of hype around generative modelling in the machine learning sense of the term – **think text and image generation.**

In contemporary applied statistics the emphasis is a bit different. Here, generative modelling refers primarily to a means of interpreting data – **the model tells a story of how a given dataset was generated.**

DAG
Directed Acyclical Graph



$$\theta \sim f(\mu)$$

Telling Generative Stories

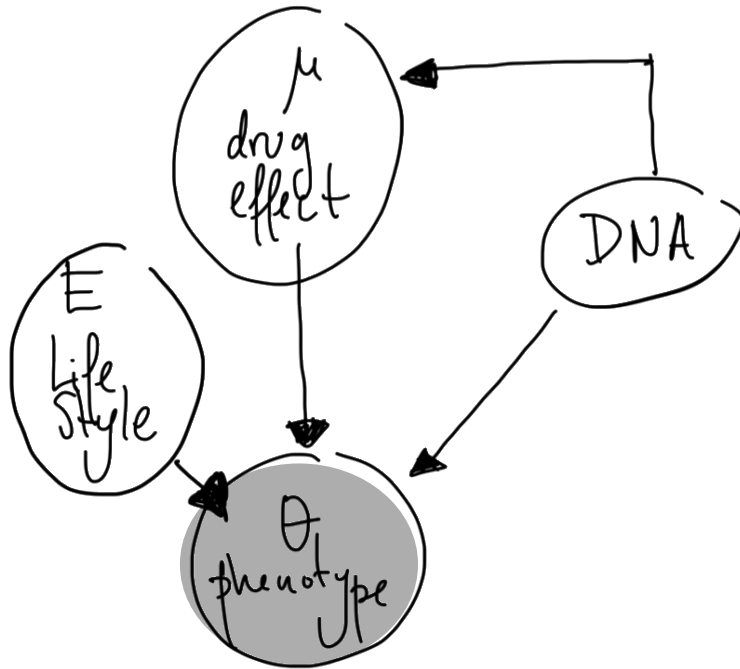
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A nice way to visualise this is via a DAG.

Example: model the likelihood of having a disease given blood biomarkers.

DAG
Directed Acyclical Graph



$$\theta \sim f(\mu, E, \text{DNA})$$

$$\mu \sim f(\text{DNA})$$

Telling Generative Stories

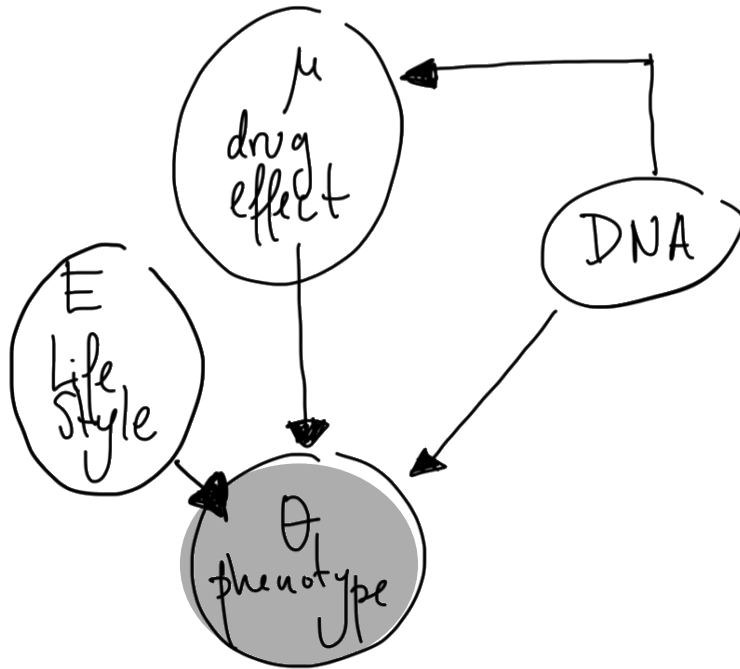
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Example: model the likelihood of having a disease given blood biomarkers.

Introducing inductive bias in the model



$$\theta \sim f(\mu, E, \text{DNA})$$

$$\mu \sim f(\text{DNA})$$

- **Dependencies**

The *graph structure*: which variables are even allowed to influence which others.

- **Functional relationships**

The *parametric form*: given that two variables are connected, what family of functions describes that relationship.

- **Priors**

The initial beliefs about the values of the parameters within that chosen family.

Bayes' Theorem Rule

and how to update beliefs based on new evidence

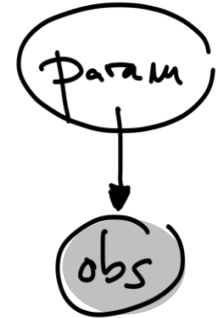
Consider observations generated by a model with some parameters.

The complete system is described by $p(\text{obs}, \text{param})$ but notice:

$$p(\text{obs}, \text{param}) = p(\text{obs}|\text{param})p(\text{param}) = p(\text{param}|\text{obs})p(\text{obs})$$

So

$$p(\text{param}|\text{obs}) = \frac{p(\text{obs}|\text{param})p(\text{param})}{p(\text{obs})}$$



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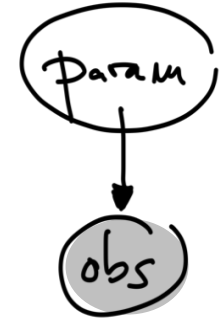
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Prior Probability: it represents our initial beliefs or knowledge about the parameters before observing any data.



Bayes' Theorem Rule

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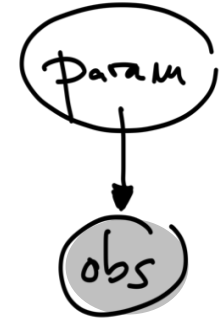
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Likelihood: it quantifies how likely the observed data is under different parameter values

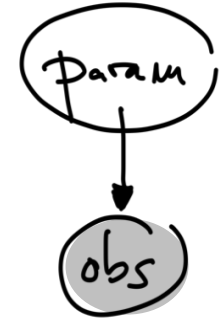
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$$\underbrace{p(\text{param}|\text{obs})}_{\text{Posterior Probability}} = \frac{\overbrace{p(\text{obs}|\text{param})}^{\text{Likelihood}} \overbrace{p(\text{param})}^{\text{Prior Probability}}}{p(\text{obs})}$$

Posterior Probability: it represents our updated beliefs about the parameters after considering the observed data.

Prior Probability: it represents our initial beliefs or knowledge about the parameters before observing any data.

Likelihood: it quantifies how likely the observed data is under different parameter values

In summary: How to update beliefs on model parameters based on new evidence

- In the absence of data, beliefs are encoded in the **prior** – which can be more or less informative.
- New observations update these beliefs via **Bayes' rule**.
- The result is **a posterior**: a balance between prior assumptions and evidence. The more informative the prior, the more data is needed to meaningfully shift it.

Example: Modelling a possibly biased coin



Example: Modelling a possibly biased coin

Model assumptions & parameters

Setting: We observe the coin being thrown in the air n times and for each instance we record if the coin lands with heads facing up, or tails. **After n throws, we record y .**

To model this system we make some **assumptions**. The key ones are:

1. The result of each throw is statistically independent.
2. The conditions are not changing from one throw to the next.

In other words, we assume **the throws are "independent and identically distributed" (iid)**.

The simplest model has **one parameter**, the latent fairness of the coin λ such that for each throw y , where $y = \text{tails}$ or $y = \text{heads}$

$$p(y = \text{tails}) = \lambda$$



Example: Modelling a possibly biased coin

Question: If after n throws we record m tails, what is the probability of having tails in the next throw? In other words, what is the estimation for λ ?

$$p(y = \text{tails}) = \lambda$$

n total throws

m total tails

Example: Modelling a possibly biased coin

Question: If after n throws we record m tails, what is the probability of having tails in the next throw? In other words, what is the estimation for λ ?

Intuitively, my estimation for the probability of tails is just the frequency of tails

$$p(y_{n+1} = \text{tails}) = \frac{m}{n}$$

$$p(y = \text{tails}) = \lambda$$

n total throws
 m total tails

Example: Modelling a possibly biased coin

Attempt 1: A classic (frequentist) approach

What is the **likelihood** of the sequence $D = \{y_1, y_2 \dots, y_n\}$ if after n throws we record m tails?

$$p(y = \text{tails}) = \lambda$$

n total throws

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Example: Modelling a possibly biased coin

Attempt 1: A classic (frequentist) approach

What is the **likelihood** of the sequence $D = \{y_1, y_2 \dots, y_n\}$ if after n throws we record m tails?

If the probability of tails is λ then the probability of heads is $1 - \lambda$

This means the probability of observing the sequence

$$p(D|\lambda) = \prod_i^n p(y_i)$$

is simply

$$p(D|\lambda) = \lambda^m (1 - \lambda)^{n-m}$$

$$p(y = \text{tails}) = \lambda$$

n total throws

m total tails

Example: Modelling a possibly biased coin

Attempt 1: A classic (frequentist) approach

What is the maximum likelihood solution for λ

$$p(D|\lambda) = \lambda^m (1 - \lambda)^{n-m}$$

One way to do this is to first take the log of the expression, then differentiate w.r.t λ , then find when the resulting expression vanishes. This gives

$$\lambda_{\text{ML}} = \frac{m}{n}$$

This is great! This is what we guessed before doing any algebra. The probability of getting a tails is given by the empirical frequency of tails in the dataset D .

$$p(y = \text{tails}) = \lambda$$

n total throws
 m total tails

Example: Modelling a possibly biased coin

Attempt 1: A classic (frequentist) approach

Question: Suppose the first flip is tails. What is the probability that the next flip will be tails?

$$p(y = \text{tails}) = \lambda$$

n total throws

m total tails

Example: Modelling a possibly biased coin

Attempt 1: A classic (frequentist) approach

Question: Suppose the first flip is tails. What is the probability that the next flip will be tails?

Even though we assume the data is iid, based on our maximum likelihood estimation of λ ,

$$\lambda_{\text{ML}} = \frac{m}{n}$$

if we observe one tails then $m=n=1$, so the probability that the next throw lands with tails facing up is exactly 1!

What went wrong?

$$p(y = \text{tails}) = \lambda$$

n total throws
 m total tails

Example: Modelling a possibly biased coin

Attempt 2: A Bayesian perspective

The problem is that the above argument **does not reflect our beliefs** about the coin.

Consider now what happens if you take a Bayesian approach with **a prior for λ**

For example, a Beta distribution:

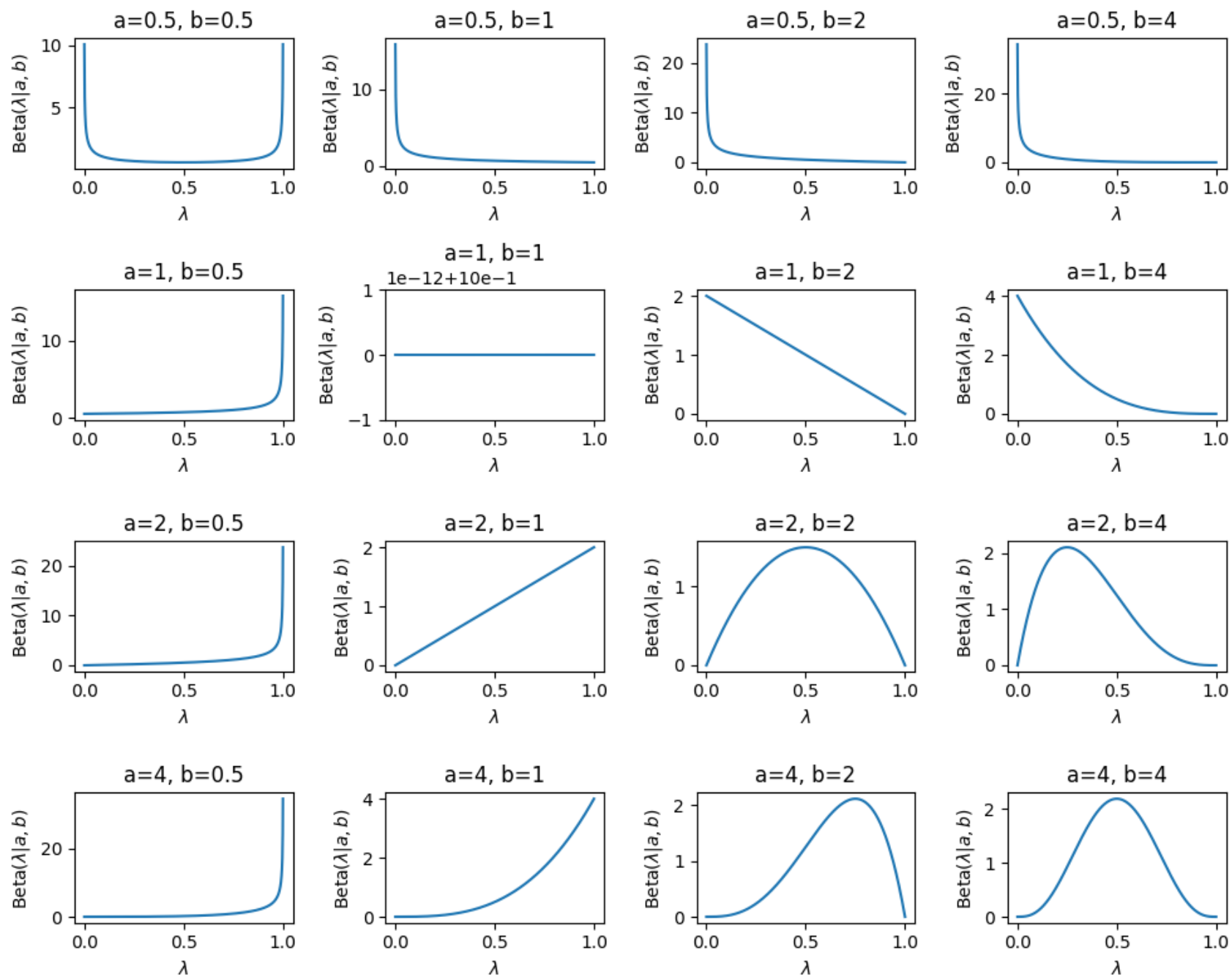
$$\text{Beta}(\lambda; a, b) = C(a, b) \lambda^{a-1} (1 - \lambda)^{b-1}$$

A very flexible, and convenient (we will explain why in a second) function for the prior.

$$p(y = \text{tails}) = \lambda$$

n total throws

m total tails



Example: Modelling a possibly biased coin

Attempt 2: A Bayesian perspective

So now we can try to model this system again, but this time using a Bayesian approach.

Computing the posterior:

Bayes rule tells us that

$$p(\lambda|D) \propto p(D|\lambda)p(\lambda)$$

$$\begin{aligned} \text{with } p(D|\lambda) &= \lambda^m (1 - \lambda)^{n-m} \\ p(\lambda) &= \text{Beta}(\lambda; a, b) \\ &= C(a, b) \lambda^{a-1} (1 - \lambda)^{b-1} \end{aligned}$$

$$p(y = \text{tails}) = \lambda$$

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 m total tails

Example: Modelling a possibly biased coin

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$$p(\lambda) = \text{Beta}(\lambda; a, b)$$

$$= C(a, b) \lambda^{a-1} (1 - \lambda)^{b-1}$$

and hence

$$p(\lambda|D) = \text{Beta}(\lambda; m + a, n - m + b)$$

$$p(y = \text{tails}) = \lambda$$

n total throws
 m total tails

Example: Modelling a possibly biased coin

Attempt 2: A Bayesian perspective

Computing the probability of tails next flip:

In Bayesian statistics, this is the probability of flipping tails conditional on the observed data $D = \{y_1, y_2, \dots, y_n\}$ which we obtain by marginalising over λ . We call this the posterior predictive.

$$p(y = \text{tails}) = \lambda$$

n total throws

m total tails

Example: Modelling a possibly biased coin

Attempt 2: A Bayesian perspective

Computing the probability of tails next flip:

In Bayesian statistics, this is the probability of flipping tails conditional on the observed data $D = \{y_1, y_2, \dots, y_n\}$ which we obtain by marginalising over λ . We call this the **posterior predictive**.

In other words

$$\begin{aligned} p(y_{n+1} = \text{tails} | D) &= \int \underbrace{p(y_{n+1} = \text{tails} | \lambda)}_{\text{posterior}} p(\lambda | D) d\lambda \\ &= \int \overline{\lambda} p(\lambda | D) d\lambda && \text{Expectation value of } \lambda \\ &= \frac{m + a}{n + a + b} \end{aligned}$$

And we see that parameters a and b play the role of pseudo counts!

$$p(y = \text{tails}) = \lambda$$

n total throws
 m total tails

Example: Modelling a possibly biased coin

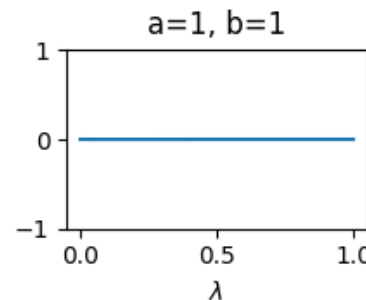
Attempt 2: A Bayesian perspective

$$p(y_{n+1} = 1|D) = \int \lambda p(\lambda|D) d\lambda = \frac{m + a}{n + a + b}$$

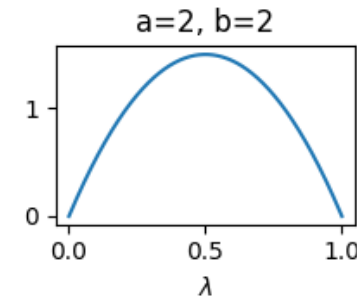
So we see that the parameters a and b play the role of pseudo counts!

More informative priors

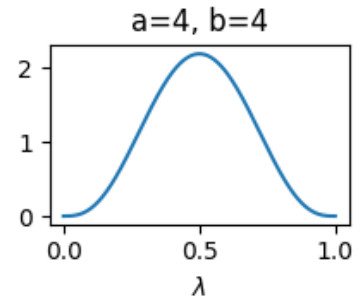
$$p(y_2 = 1|y_1 = 1) = \frac{1 + a}{1 + a + b}$$



$$2/3 = 0.667$$

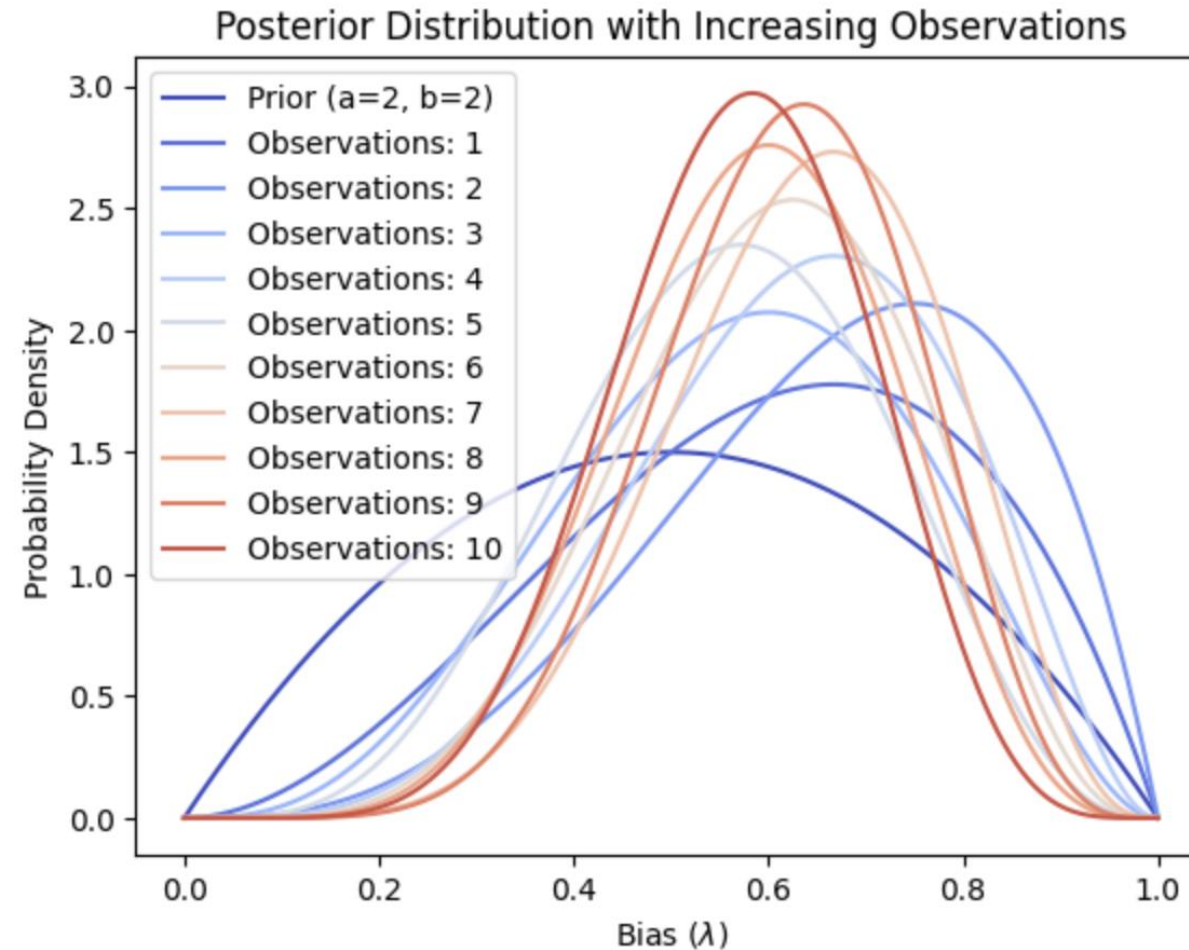


$$3/5 = 0.6$$



$$5/9 = 0.556$$

Updating beliefs upon new observations

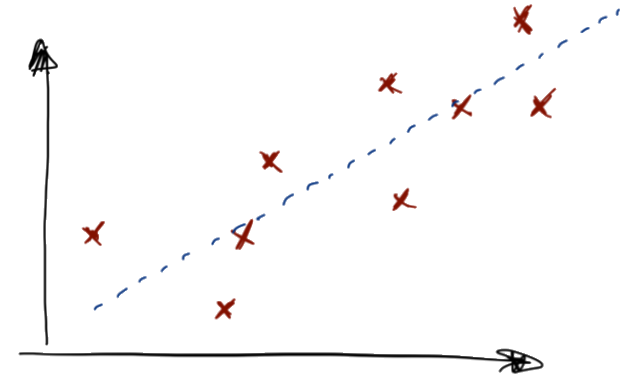


Examples of inductive biases you've probably already faced: Smoothness and Regularisation

In biology, **data is often noisy** — making **smoothness** a natural and well-motivated assumption.

How to inject this assumption in model building:

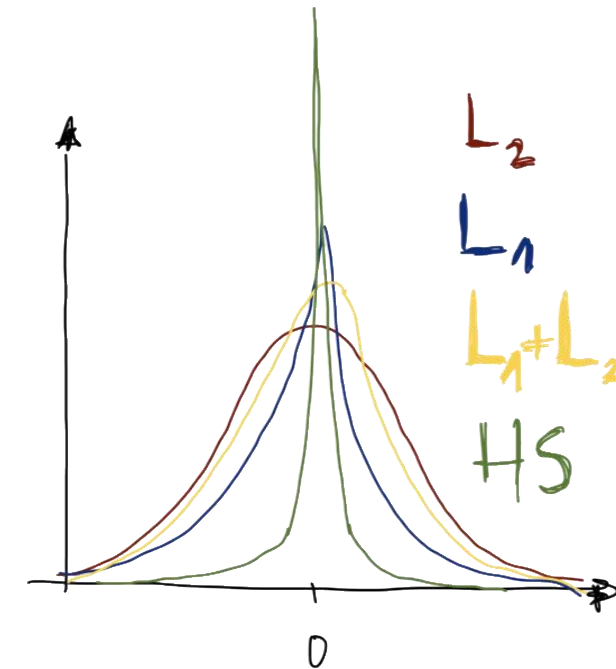
- **Functional form:** choosing a simple model (e.g. linear regression) encodes smoothness directly — linearity rules out wild oscillations by construction.
- **Parameter priors:** in more complex models, penalise parameters that would produce rough functions – **regularisation**. Alternatively, smoothness itself can be treated as a learnable parameter with a prior that favours regularity.



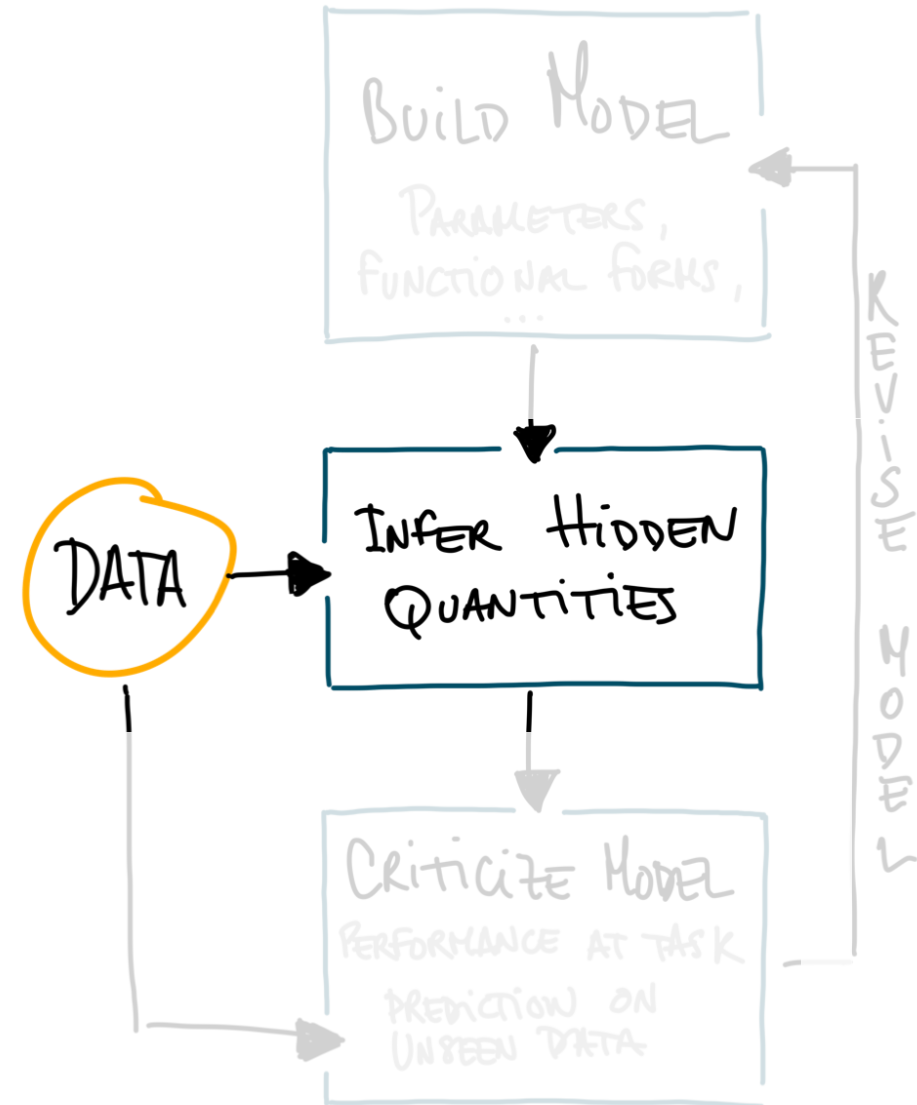
Examples of inductive biases you've probably already faced: Smoothness and Regularisation

Regularisation is a very powerful tool in biology. You might have dealt with it but not realised it maps to parameter priors:

Regularisation	Bayesian prior	Effect
L2 (Ridge)	Gaussian	Small, distributed weights
L1 (Lasso)	Laplace	Sparse weights — many exactly zero
L1 + L2 (Elastic Net)	Gaussian + Laplace	Sparsity + stability for correlated variables
Horseshoe	Horseshoe	Shrinks noise aggressively, leaves large signals intact



Two notes on inference

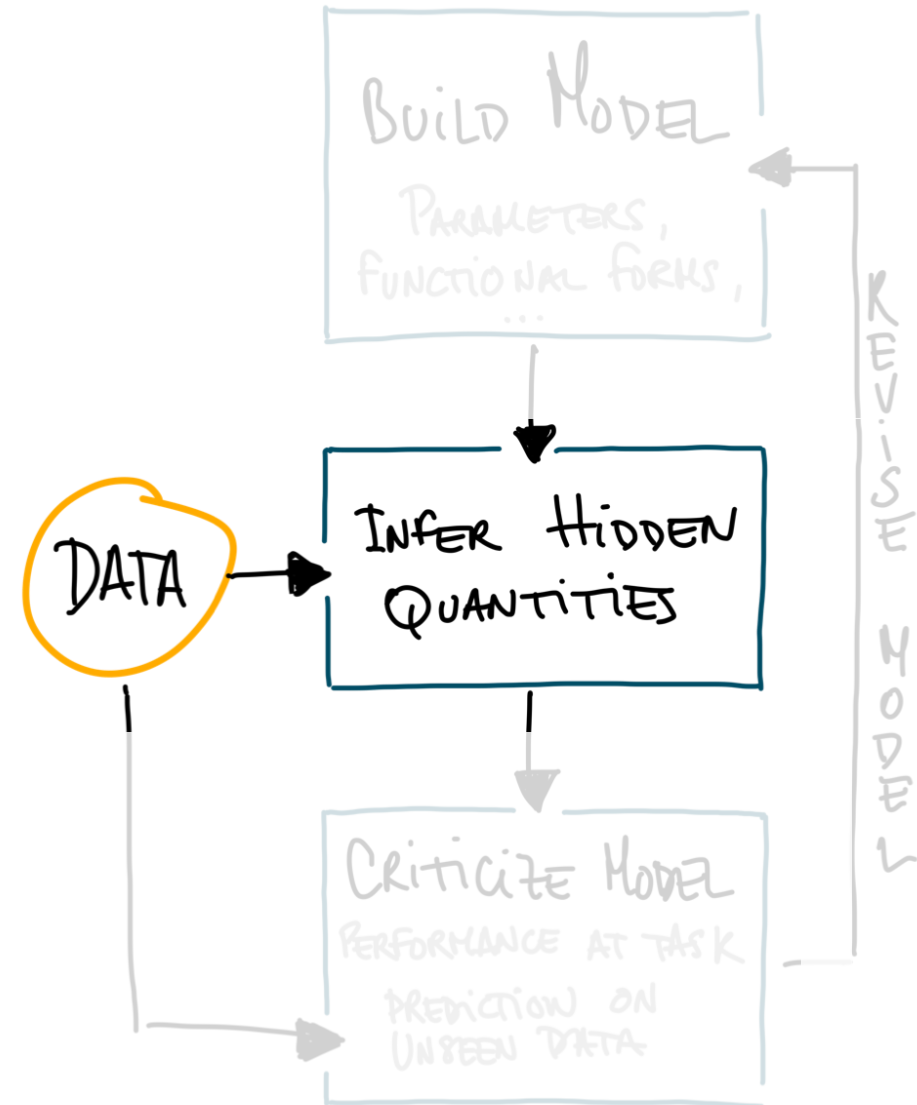


Two notes on inference – 1. Conjugate priors

Bayes' rule:

$$\underbrace{p(z|x)}_{\text{How the data determines your model}} = \frac{\overbrace{p(x|z)p(z)}^{\text{"our model"} \quad \underbrace{\text{prior belief}}}{p(x)}$$

How the data
determines your model



Two notes on inference – 1. Conjugate priors

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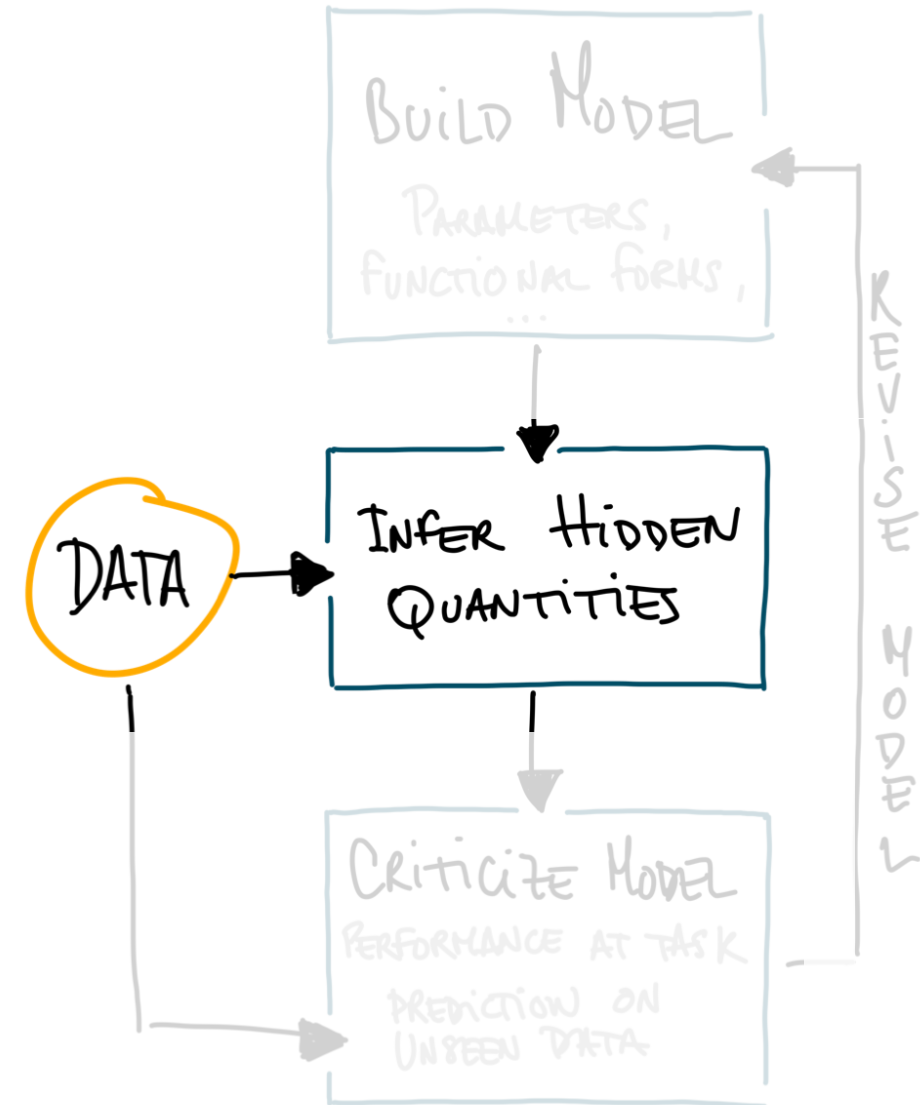
In the coin flipping example:

Prior: $p(\lambda) = \text{Beta}(\lambda; a, b)$

Posterior: $p(\lambda|D) = \text{Beta}(\lambda; m + a, n - m + b)$

Whenever the prior distribution is of the same family as the posterior, it is called a **conjugate prior** for the likelihood function.

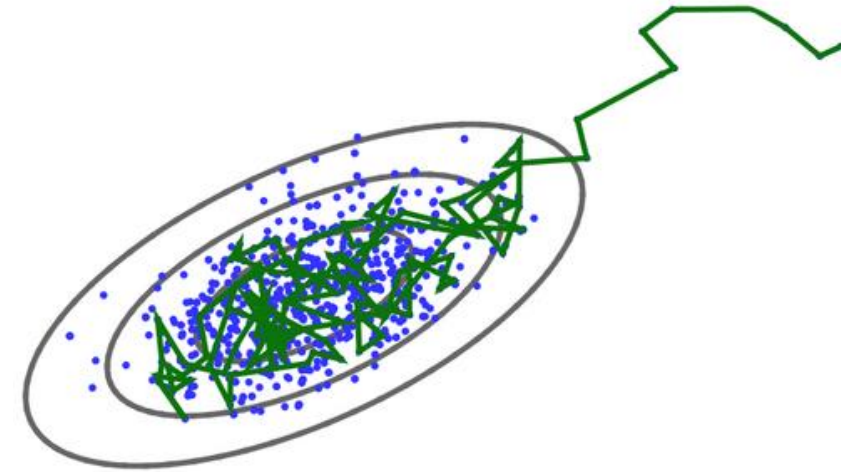
└─ We know the functional form of the posterior!



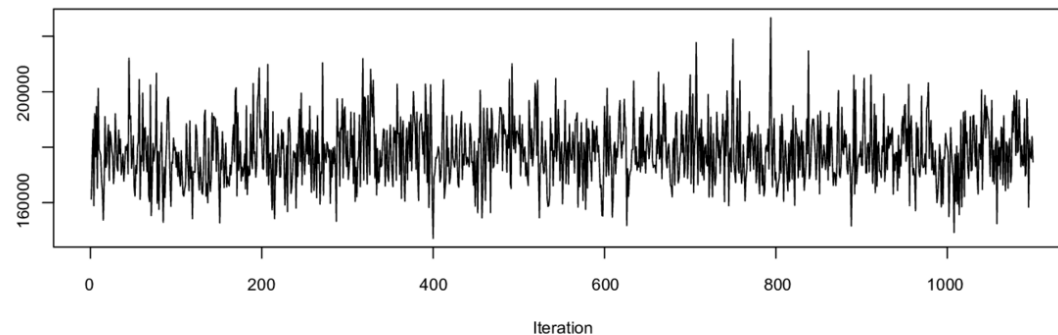
Two notes on inference – 2. Numerical approaches

MCMC: exploring the posterior with a Markov chain

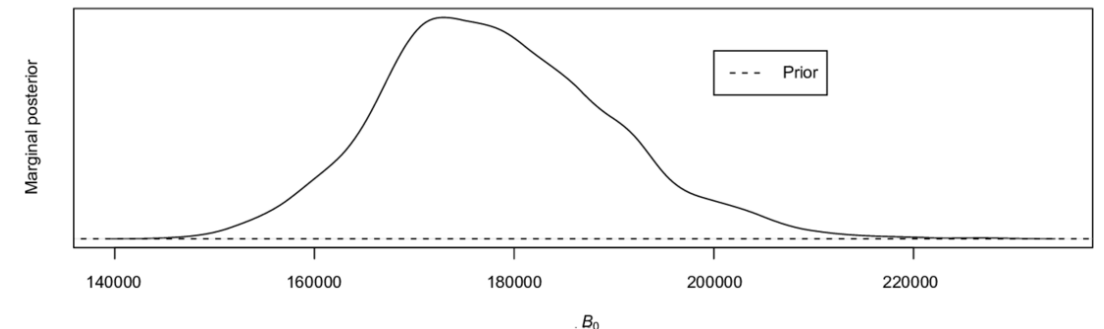
1. The algorithm starts with an initial guess for the parameter.
2. It then proposes a new value for the parameter, based on a proposal distribution.
3. It attributes a probability based on Bayes' rule.
4. In the large sample limit it converges to the posterior.



Trace plot



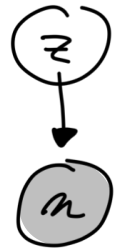
Sampled posterior



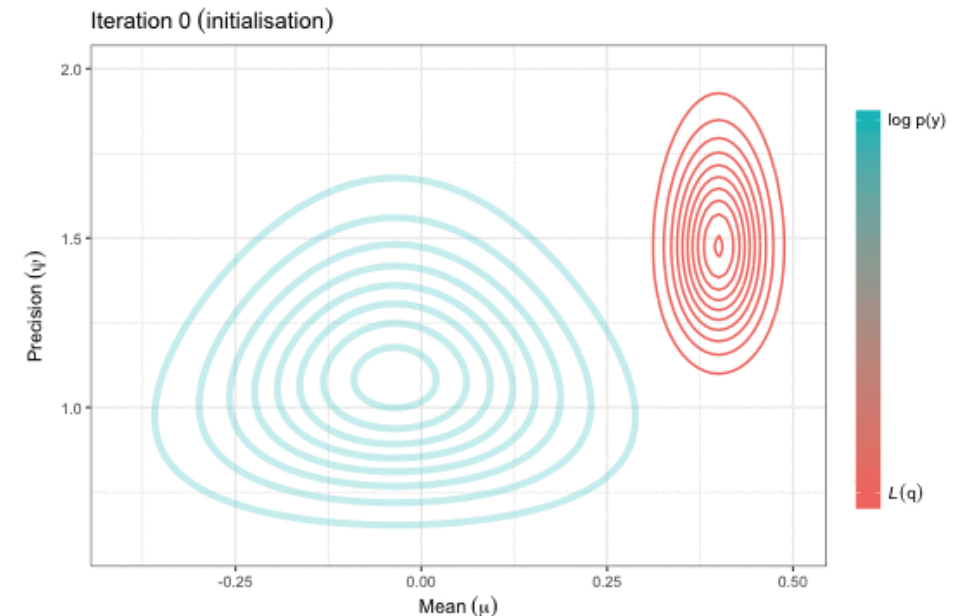
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Variational Inference: approximating the posterior with a guide distribution

1. Choose a guide distribution $q_\phi(z)$
2. Find the best parameters ϕ that minimise the distance (KL divergence) between $q_\phi(z)$ and the posterior $p(z|x)$



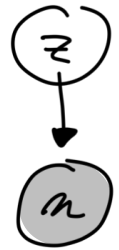
$$\phi^* = \arg \min_{\phi} D_{\text{KL}}(q_{\phi}(\theta) \parallel p(\theta \mid x))$$



Two notes on inference – 2. Numerical approaches

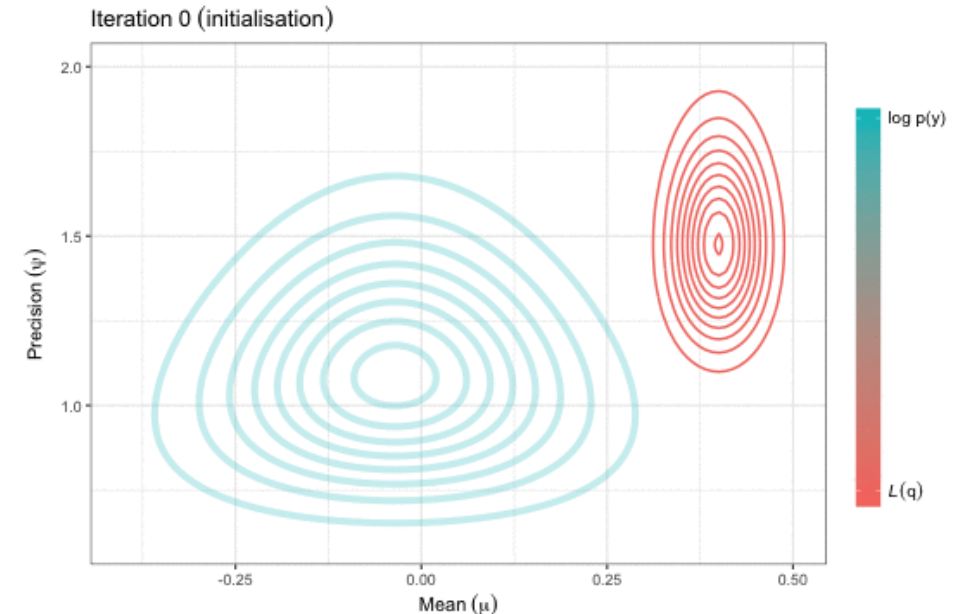
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3. In practice, minimising the KL divergence is done via maximising the Evidence Lower Bound (ELBO)



$$\phi^* = \arg \min_{\phi} D_{\text{KL}}(q_{\phi}(\theta) \parallel p(\theta \mid x))$$

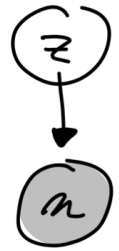
$$\mathcal{L}(\phi) = \underbrace{\mathbb{E}_{q_{\phi}}[\log p(x \mid \theta)]}_{\text{fit the data}} - \underbrace{D_{\text{KL}}(q_{\phi}(\theta) \parallel p(\theta))}_{\text{stay close to prior}}$$



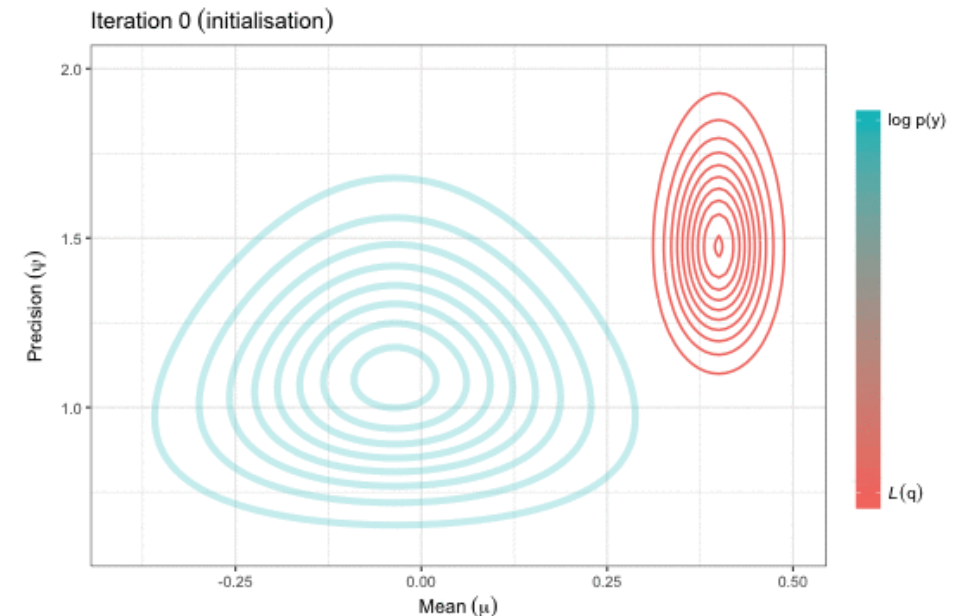
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Advantages of MCMC Vs VI – Accuracy vs. Speed Trade-off



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